

The Digital Consumer: Utility, Demand, and Elasticity

ITGE275 Economics

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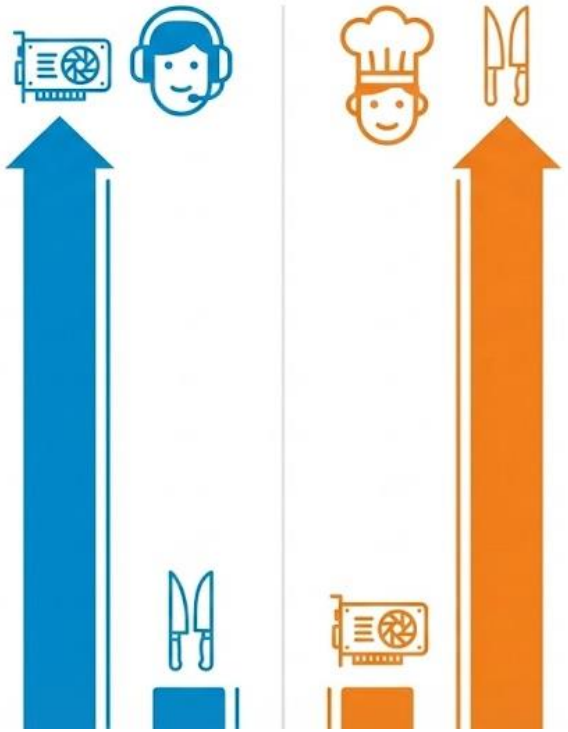
The Consumer as an Autonomous Agent

- Last week, we looked at the Firm (Producer) optimizing production.
- Today, we look at the **Consumer** (You).
- In CS terms, a consumer is an **autonomous agent** running a continuous optimization loop.
- **Goal:** Maximize Happiness (Utility).
- **Constraint:** Limited Resources (Budget/Income).
- **Input Data:** Market Prices.



Utility Theory: The "Satisfaction Algorithm"

- Why do we buy things? Because they provide **Utility**.
- **Utility (U):** A measure of satisfaction or happiness derived from consuming a good.
- It is the "Return Function" or "Reward Function" in Reinforcement Learning terms.
- **Note:** Utility is subjective. The utility of a GPU is high for a gamer, but low for a chef.

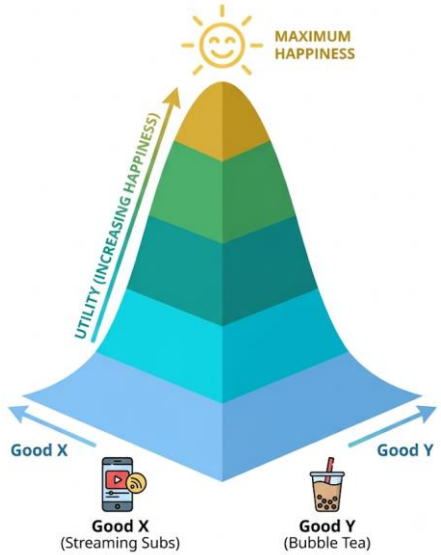


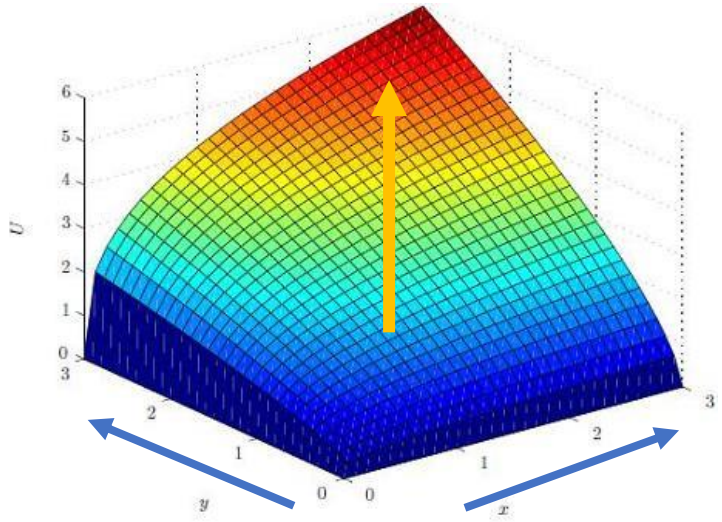
The Utility Function (The Math)

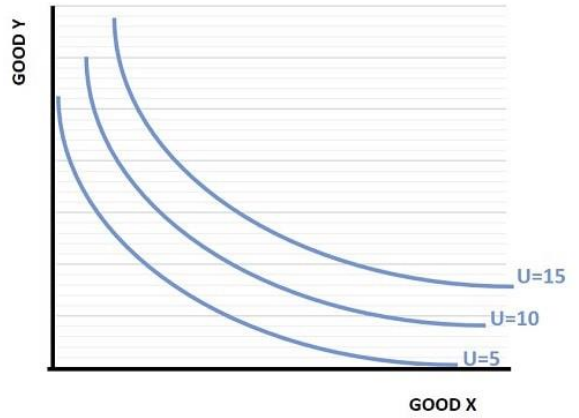
- We model preferences mathematically.
- Common Model: **Cobb-Douglas Utility Function**.

$$U(\mathrm{x}, \mathrm{y}) = x^{\alpha} \cdot y^{\beta}$$

- Where:
  - x, y: Quantities of two different goods (e.g., Streaming Subs vs. Bubble Tea).
  - α, β: Preference parameters (How much you "like" each good).
- This function creates a "hill" of happiness we want to climb.

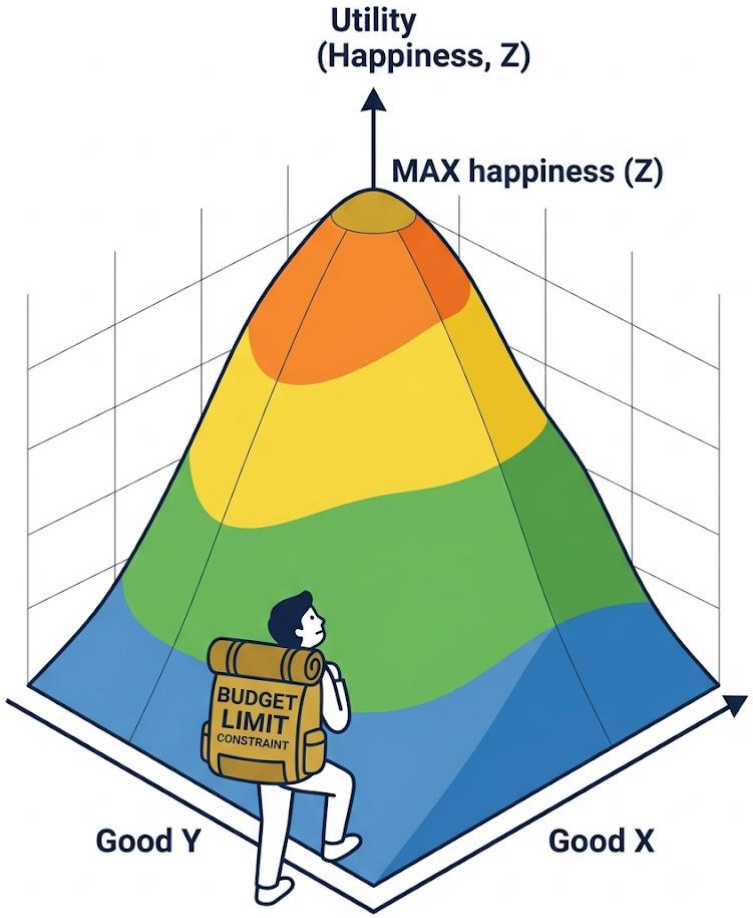






Visualizing Preferences in 3D

- In today's tutorial, we will use matplotlib to plot this 3D surface.
- The **Z-axis** represents your level of happiness.
- Your goal as a rational agent is to reach the highest Z-value possible.
- However, you are grounded by your **Budget Constraint** (you can't just buy infinite x and y).

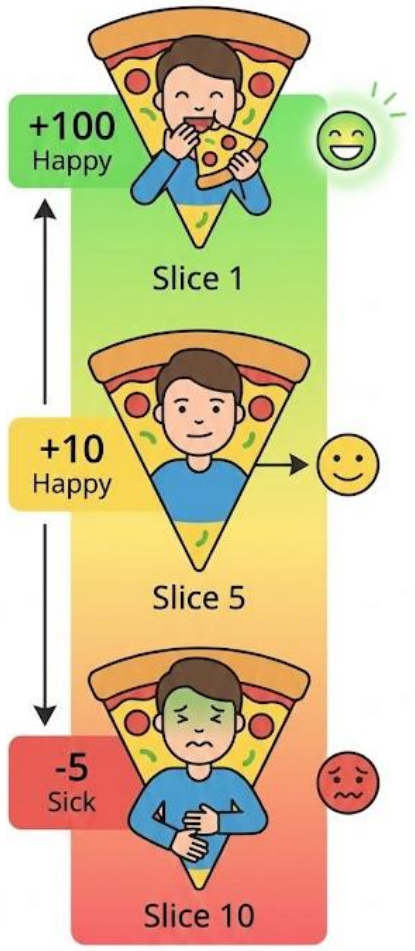


Marginal Utility

- **Marginal Utility (MU):** The *additional* satisfaction gained from consuming *one more* unit of a good.
- Mathematical definition: The partial derivative of the Utility function with respect to ***x***.

$$MU_x = \frac{\partial U}{\partial x}$$

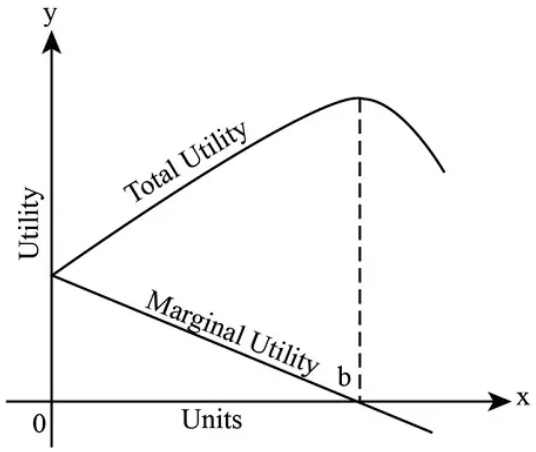
- This is the "Gradient" of the utility hill.

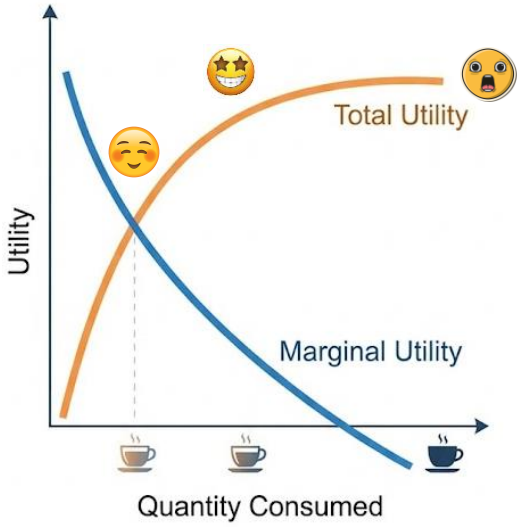


Gradient of the Utility Hill: Diminishing Satisfaction

Law of Diminishing Marginal Utility

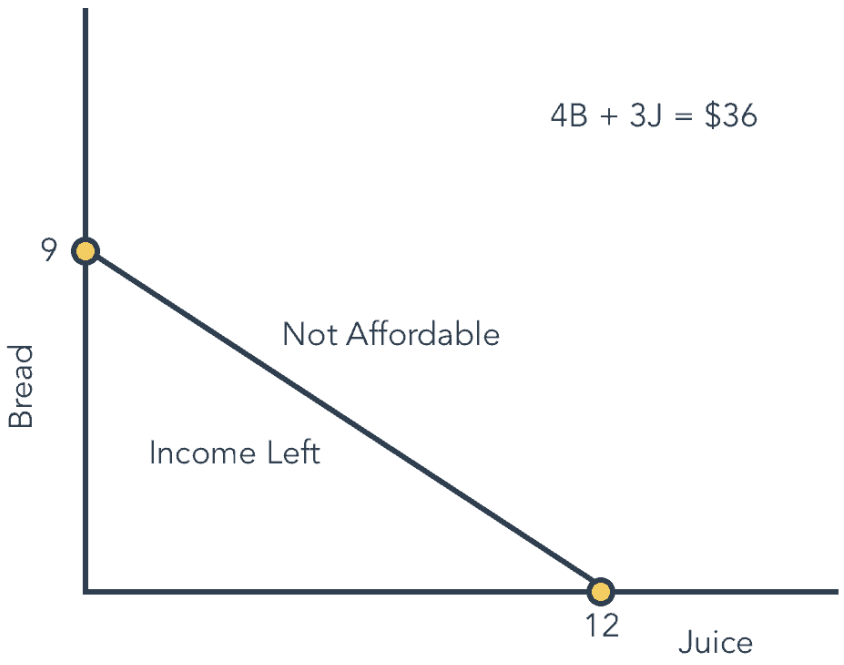
- **The Principle:** As you consume more of a good, the marginal utility of each additional unit **decreases**.
- **Example:**
  - 1st Coffee: Essential (High Utility).
  - 2nd Coffee: Nice (Medium Utility).
  - 5th Coffee: Jitters (Low/Negative Utility).
- **CS Implication:** This is why we use non-linear functions (like square roots or logs) in our models, not straight lines.





The Budget Constraint

- If Utility is the objective, the **Budget** is the constraint.
- Formula: $P_x \cdot x + P_y \cdot y \leq I$
  - P_x, P_y: Prices of goods.
  - I: Income.
- This forms a linear boundary in our 2D space of goods.
- You can only consume combinations *inside* or *on* this line.

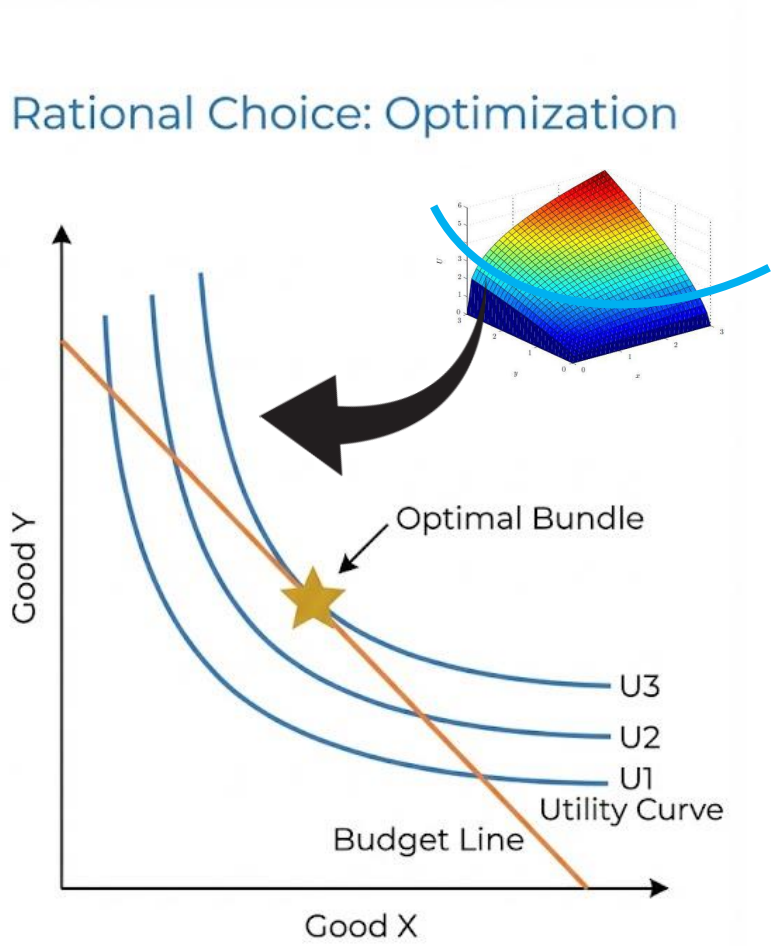


https://www.intelligenteconomist.com/budget-constraint/

The Optimization Problem Solved

- **Rational Choice:** The consumer chooses the combination of goods (x, y) that touches the highest possible Utility curve while staying within the Budget line.
- **Tangency Condition:** This happens where the "Bang for Buck" is equal for all goods.

$$\frac{MU_x}{P_x} = \frac{MU_y}{P_y}$$



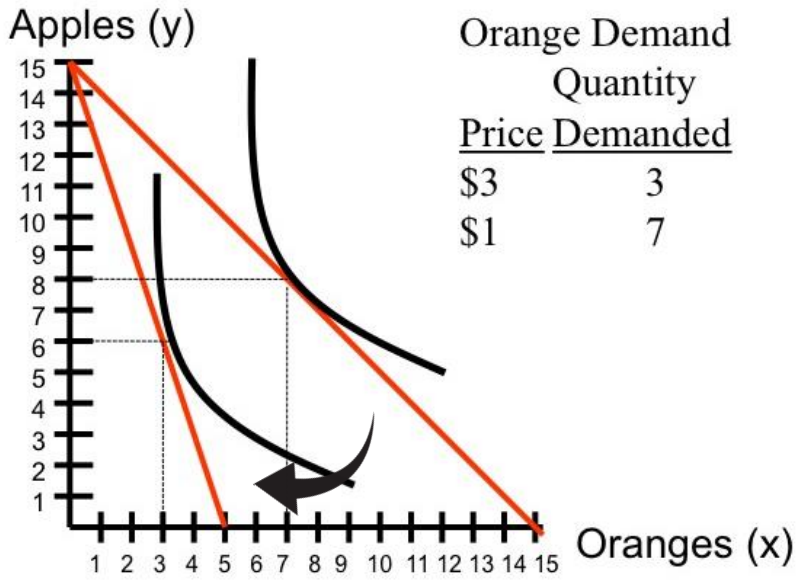
Part 1 Summary

- Consumers are optimizers.
- **Objective:** Maximize U(x, y).
- **Constraint:** Income ≥ Spending.
- **Result:** We equate the marginal utility per dollar across all goods.
- **Next:** How does changing the Price affect this decision? (Demand).



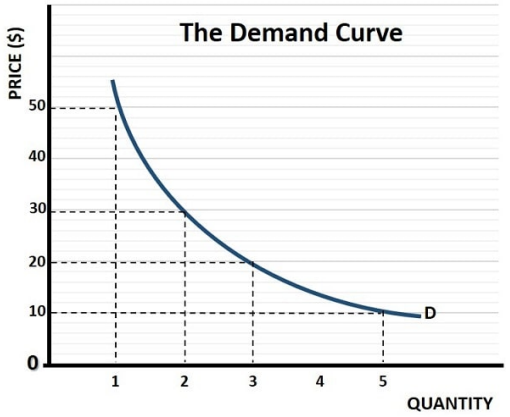
Deriving Demand

- What happens to our optimal choice if the **Price** of Good X goes up?
  - The Budget Line pivots inward (You can afford less X).
  - You are forced to a lower Utility curve.
  - You choose a new optimal point with **less X**.
- This relationship (Price ↑, Quantity ↓) is the **Demand Curve**.



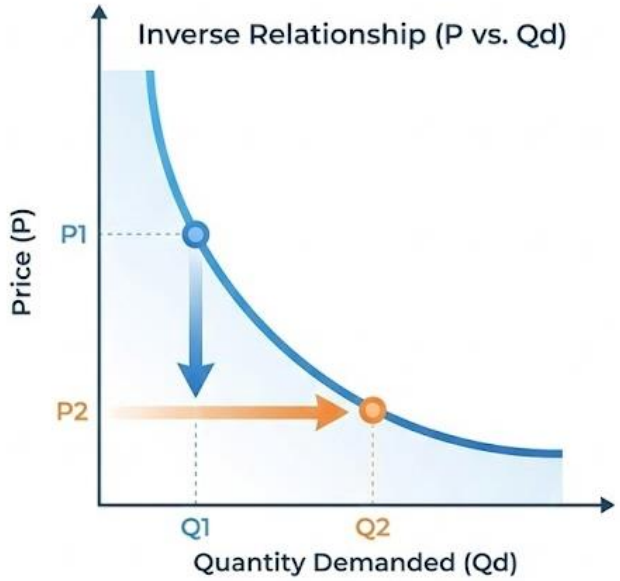
Orange Demand

Price	Quantity Demanded
$3	3
$1	7



The Law of Demand

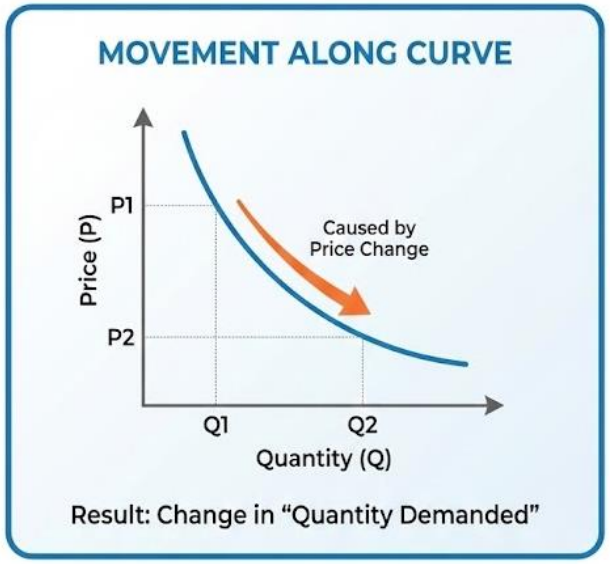
- **Law of Demand:** There is an **inverse relationship** between Price (***P***) and Quantity Demanded (Q_d), ceteris paribus (all else being equal).
- **Why?**
  - **Income Effect:** You feel poorer, so you buy less.
  - **Substitution Effect:** You switch to cheaper alternatives.

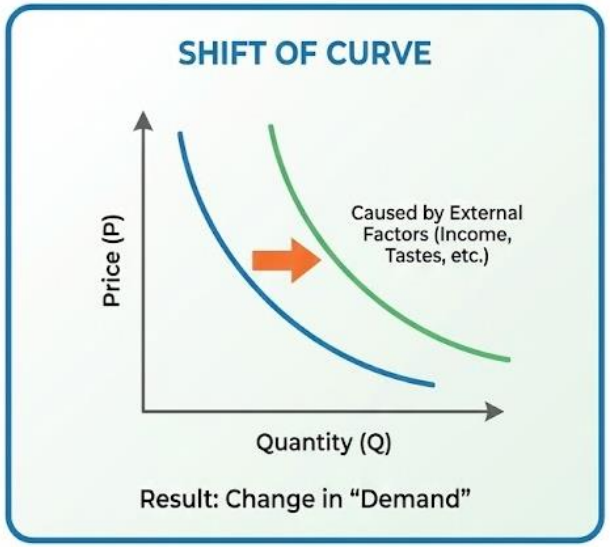




Critical Concept: Shift vs. Movement

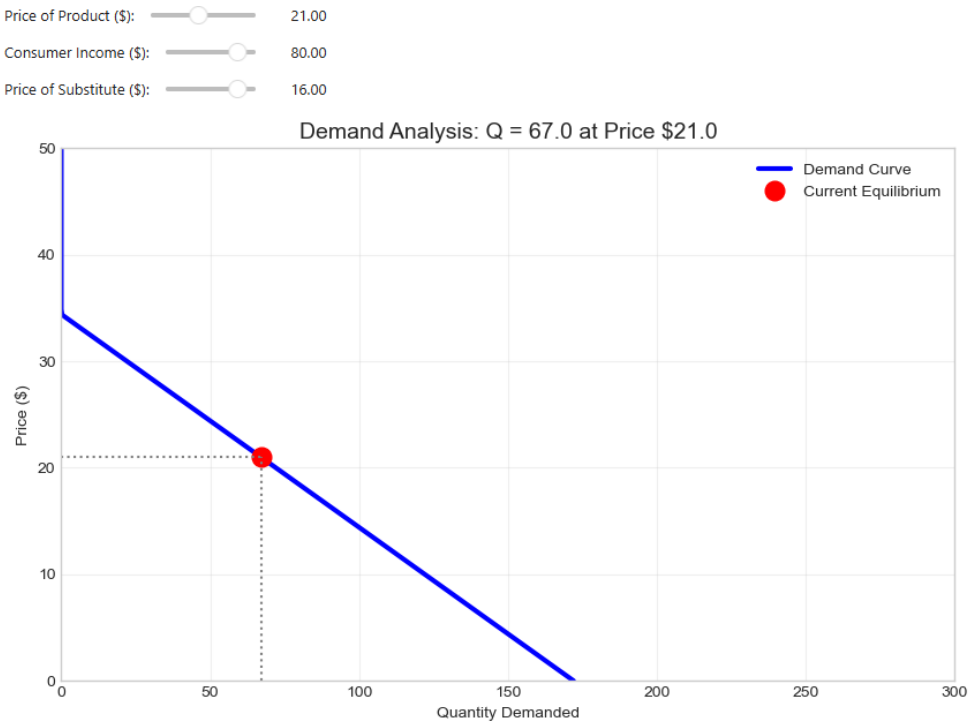
- This is the #1 confusion in economics (and exams).
- **Movement Along the Curve:**
  - Caused by: Change in **Price** of the good itself.
  - Result: Change in "Quantity Demanded."
- **Shift of the Curve:**
  - Caused by: Change in **External Factors** (Income, Tastes, Price of Substitutes).
  - Result: Change in "Demand."





Interactive Demand Curve (Tutorial Preview)

- In the Jupyter Notebook, we will code an interactive widget to visualize this.
- **Slider 1 (Price):** Moves the red dot *along* the curve.
- **Slider 2 (Income):** *Shifts* the blue curve left/right.
- **Slider 3 (Substitute Price):** *Shifts* the curve based on competition.



Shift Factor 1: Income

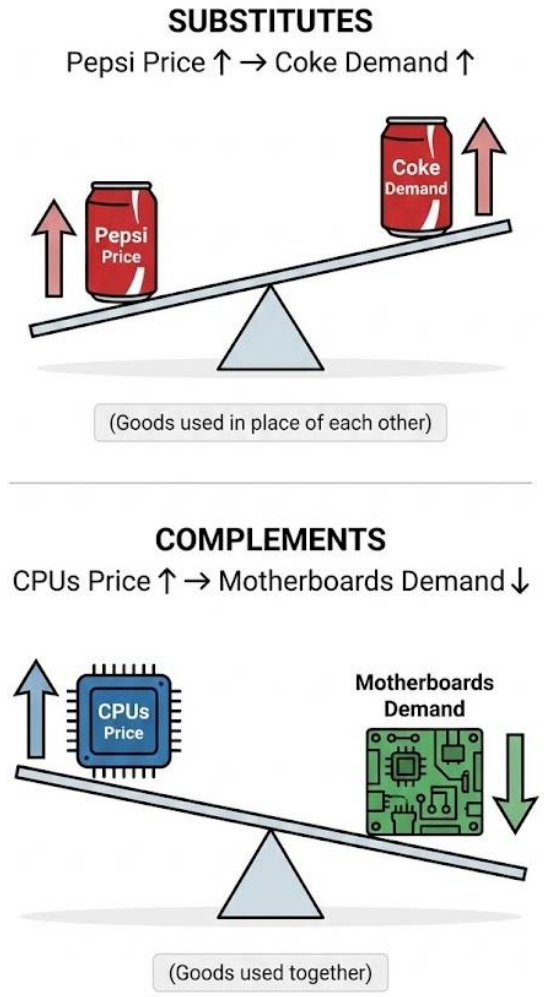
- **Normal Goods:** Income ↑ → Demand ↑ (Shift Right).
  - Example: IPhone, Sushi, Vacations.
- **Inferior Goods:** Income ↑ → Demand ↓ (Shift Left).
  - Example: Instant Noodles, Bus Tickets (You switch to a car).





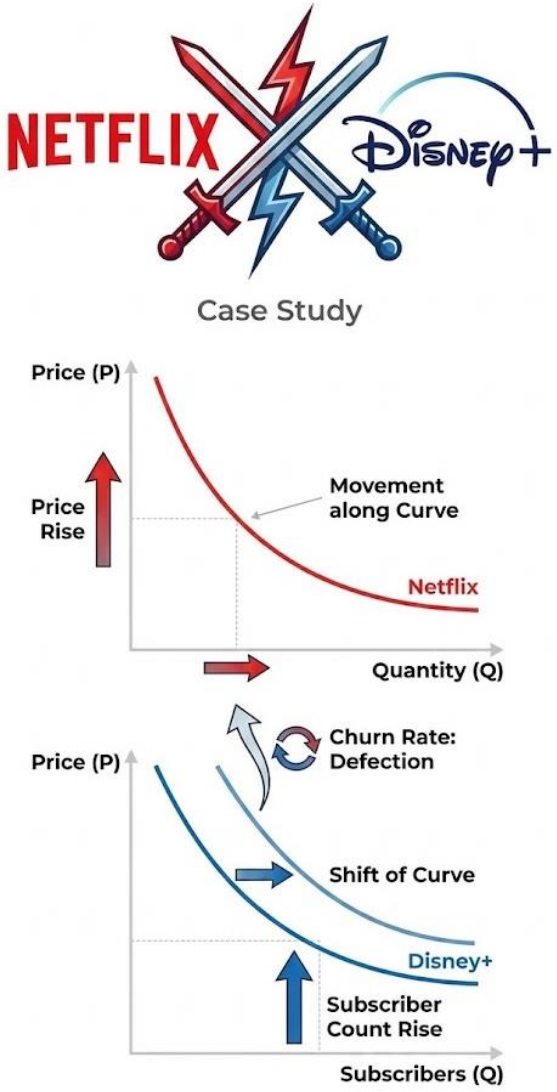
Shift Factor 2: Substitutes & Complements

- **Substitutes:** Goods used in place of each other.
  - Price of Pepsi ↑ → Demand for Coke ↑ (Shift Right).
- **Complements:** Goods used together.
  - Price of CPUs ↑ → Demand for Motherboards ↓ (Shift Left).



Case Study: Streaming Wars (Substitutes)

- **Context:** Netflix vs. Disney+.
- These are close substitutes.
- If Netflix raises prices (Movement along Netflix curve), consumers cancel and subscribe to Disney+ (Shift of Disney+ curve).
- **Churn Rate:** The metric measuring this defection.

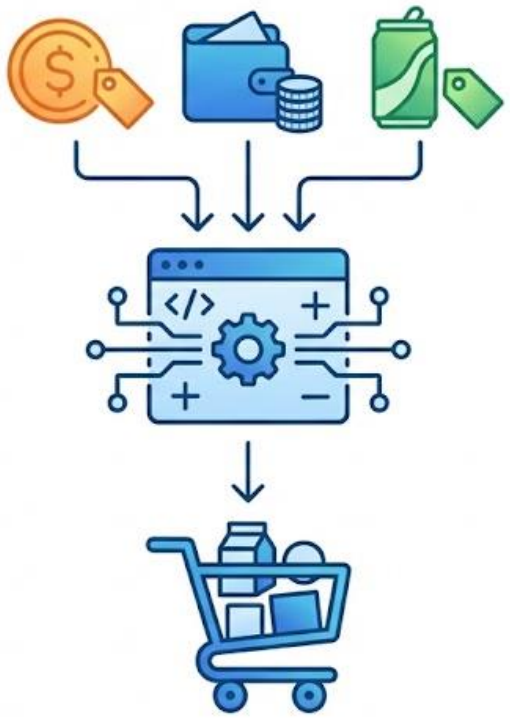


Modeling Linear Demand

- A simple computational model for demand:

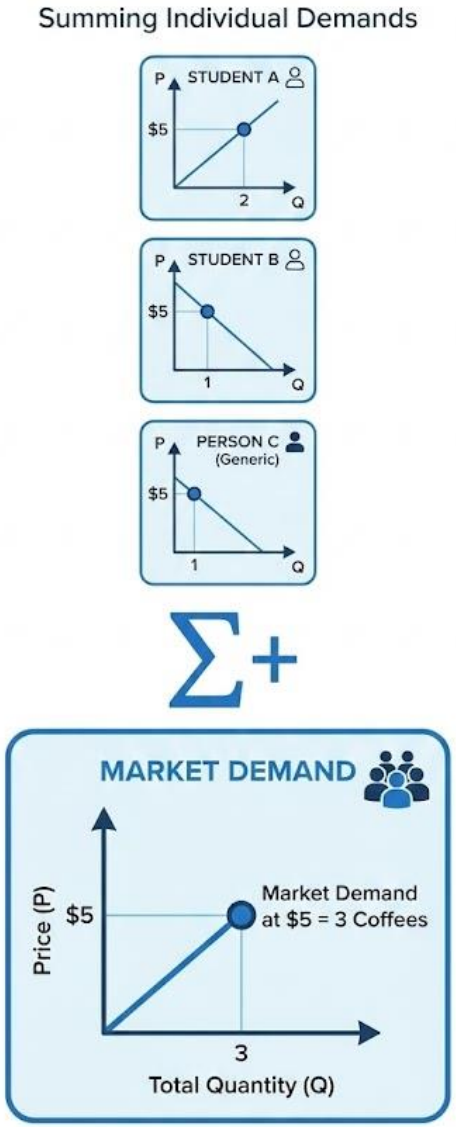
$$Q_d = A - (B \cdot P) + (C \cdot I) + (D \cdot P_{sub})$$

- In the lab, we will use Python functions to calculate $\boldsymbol{Q_d}$ dynamically based on these inputs.



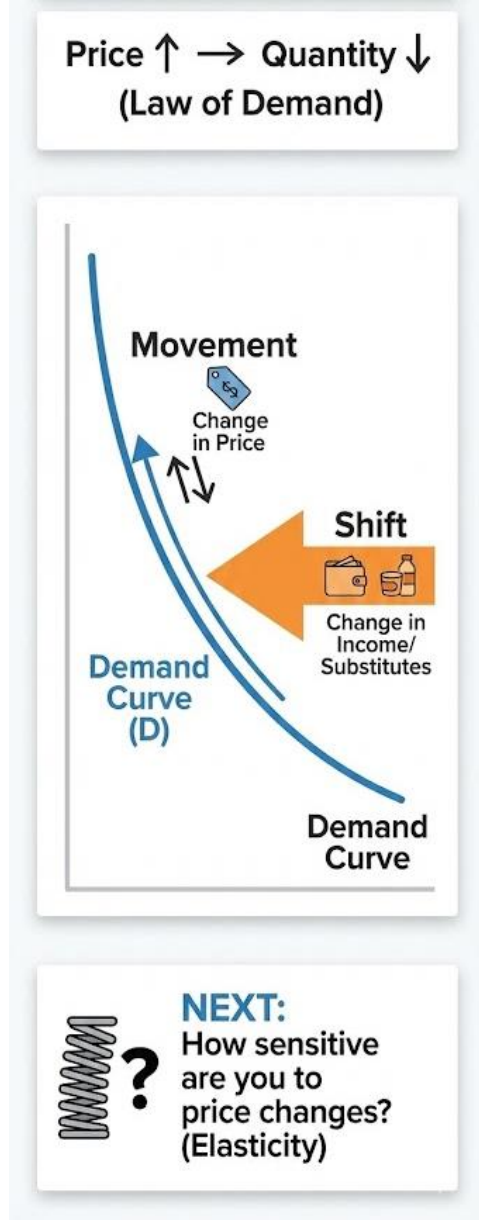
From Individual to Market Demand

- **Market Demand:** The sum of all individual demands.
- If Student A buys 2 coffees at $5, and Student B buys 1 coffee at $5...
- Market Demand at $5 = 3 coffees.
- Big Data allows tech companies to estimate this aggregate curve precisely.



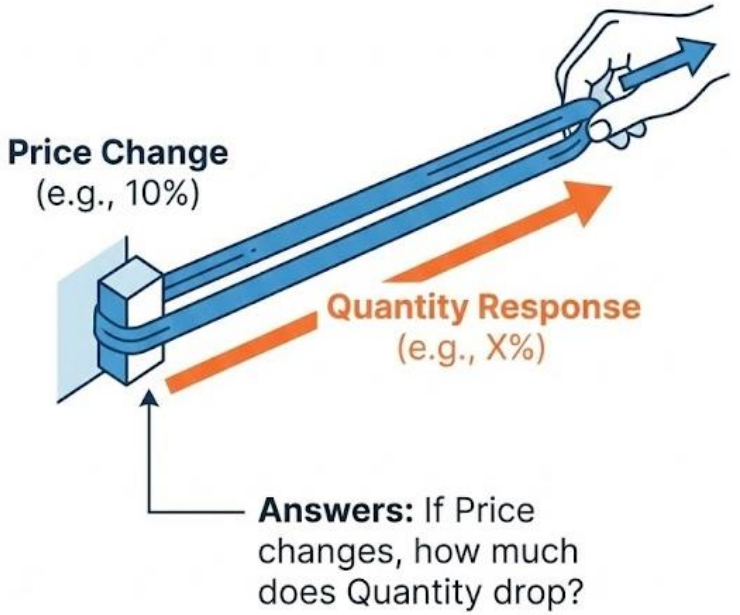
Part 2 Summary

- Price ↑ = Quantity ↓ (Law of Demand).
- Change in Price = Movement.
- Change in Income/Substitutes = Shift.
- **Next:** How *sensitive* are you to these price changes? (Elasticity).



Introduction to Elasticity

- **Elasticity:** A measure of responsiveness.
- It answers the question: "If I change the price by 10%, by what % will the quantity drop?"
- Crucial for pricing strategy and revenue management.

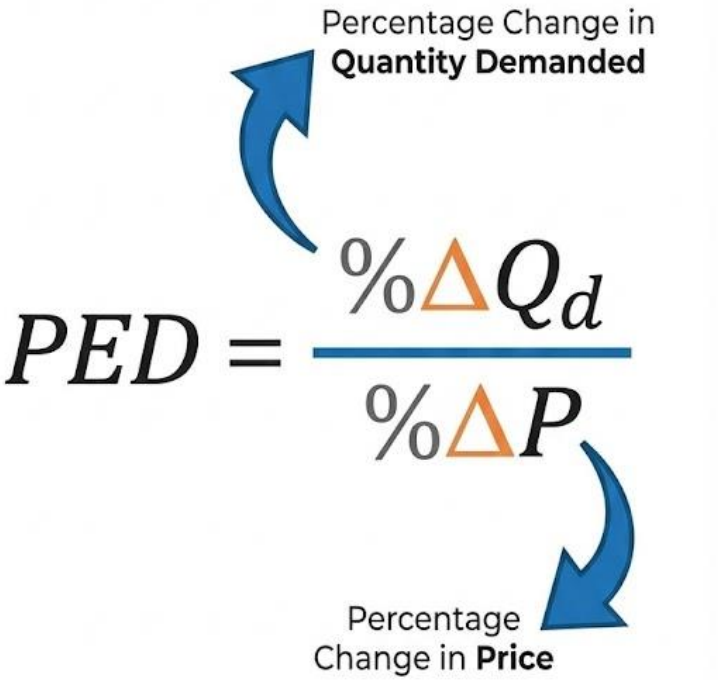


Price Elasticity of Demand (PED)

- **Formula:**

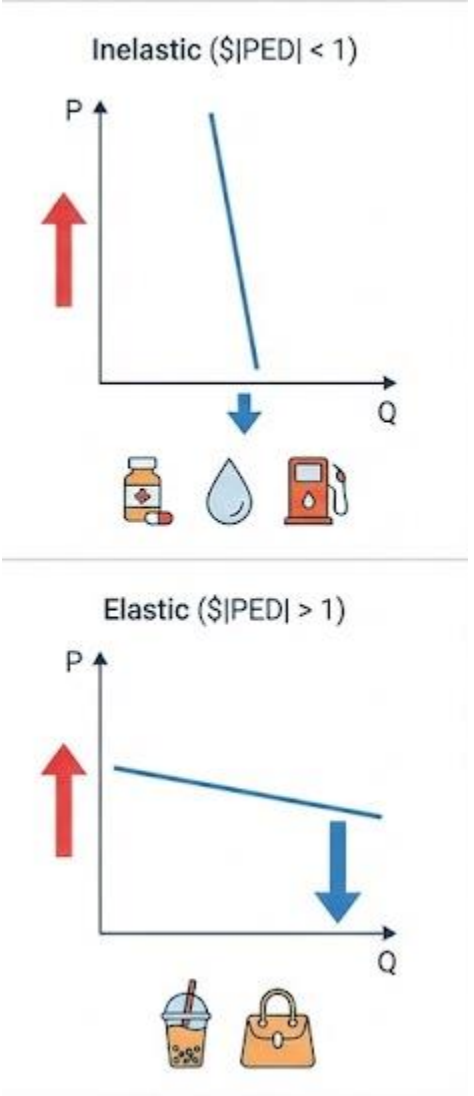
$$PED = \frac{\%\Delta Q_d}{\%\Delta P}$$

- **Interpretation:**
  - We usually take the absolute value |***PED***|.
  - It tells us the "sensitivity score."



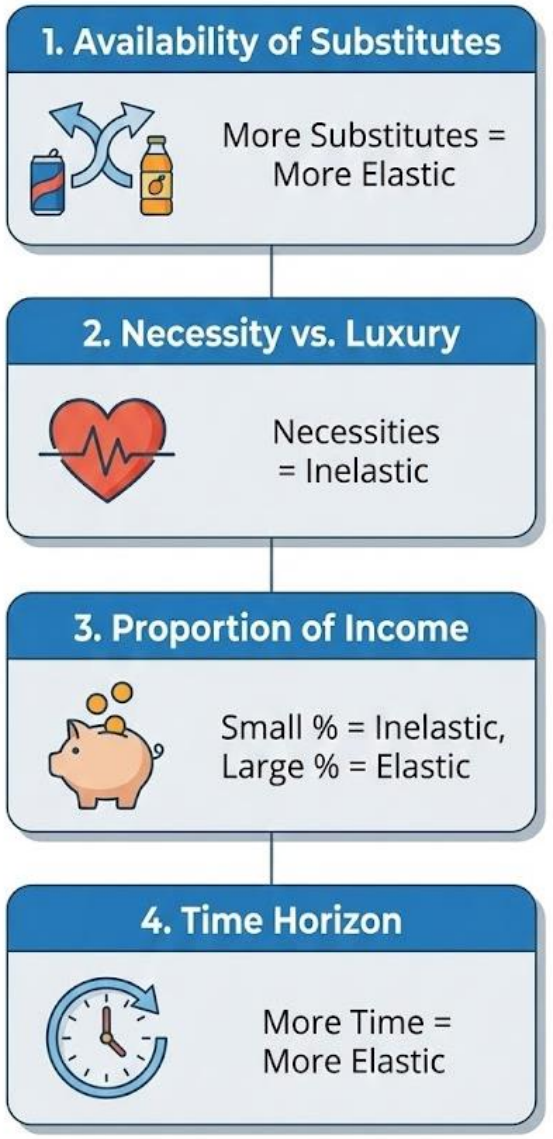
Elastic vs. Inelastic

- **Inelastic (|PED| < 1):** Not sensitive.
  - Price ↑ 10% → Quantity ↓ 2%.
  - Examples: Medicine, Water, Gasoline (Necessities).
- **Elastic (|PED| > 1):** Very sensitive.
  - Price ↑ 10% → Quantity ↓ 20%.
  - Examples: Bubble Tea, Luxury Bags, Specific Brands (Luxuries).



Determinants of Elasticity

- **1. Availability of Substitutes:** More substitutes = More Elastic. (Coke is elastic because Pepsi exists).
- **2. Necessity vs. Luxury:** Necessities are Inelastic.
- **3. Proportion of Income:** Cheap things (salt) are inelastic. Expensive things (cars) are elastic.
- **4. Time Horizon:** Demand becomes more elastic over time (people find alternatives).



Elasticity and Total Revenue

- **Total Revenue (TR) = Price x Quantity.**
- The relationship between Price and Revenue depends on Elasticity.
- This is the "Revenue Test."

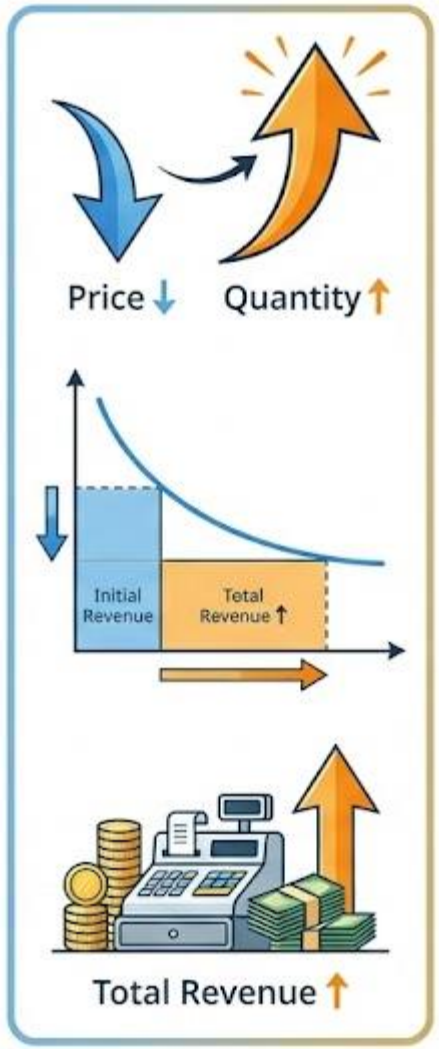


Revenue Strategy: Elastic Goods

- **Context:** Demand is Elastic (>1). Quantity drops *faster* than Price rises.
- **Strategy: Lower the Price.**
  - Price ↓ slightly → Quantity ↑ massively.
  - **Result:** Total Revenue ↑.
- **Example:** "Steam Summer Sale." Games are cheap, volume explodes, revenue hits record highs.



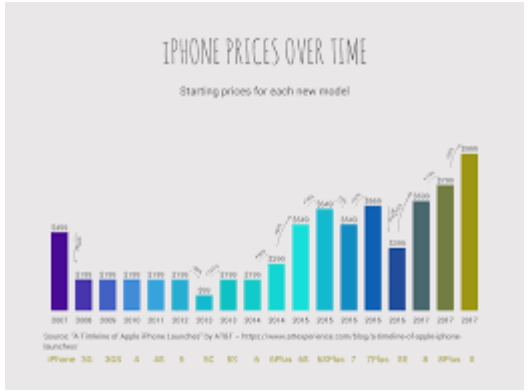


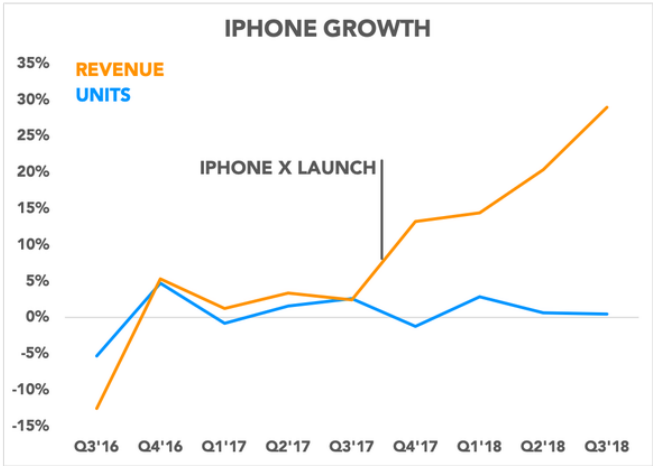


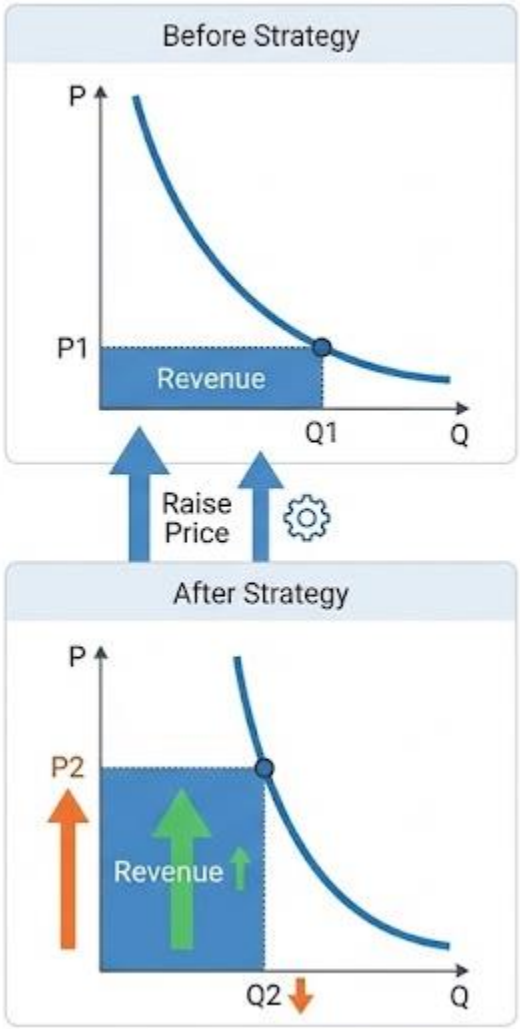
Revenue Strategy: Inelastic Goods

- **Context:** Demand is Inelastic (<1). Quantity drops *slower* than Price rises.
- **Strategy: Raise the Price.**
  - Price ↑ significantly → Quantity ↓ slightly.
  - **Result:** Total Revenue ↑.
- **Example:** Insulin, Gasoline, Apple ecosystem (high switching costs).



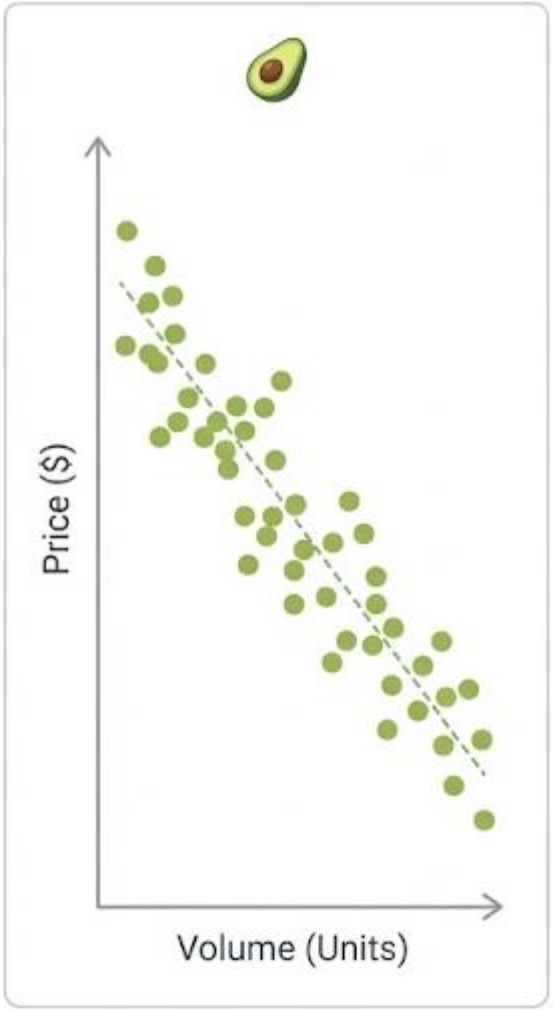






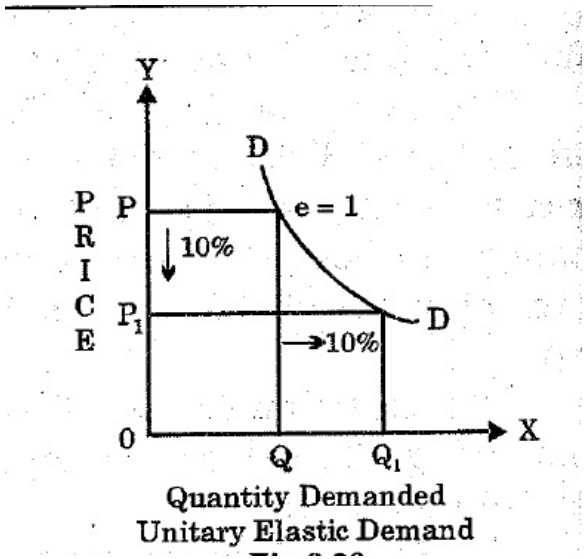
Case Study: Avocado Dataset (Synthetic Data)

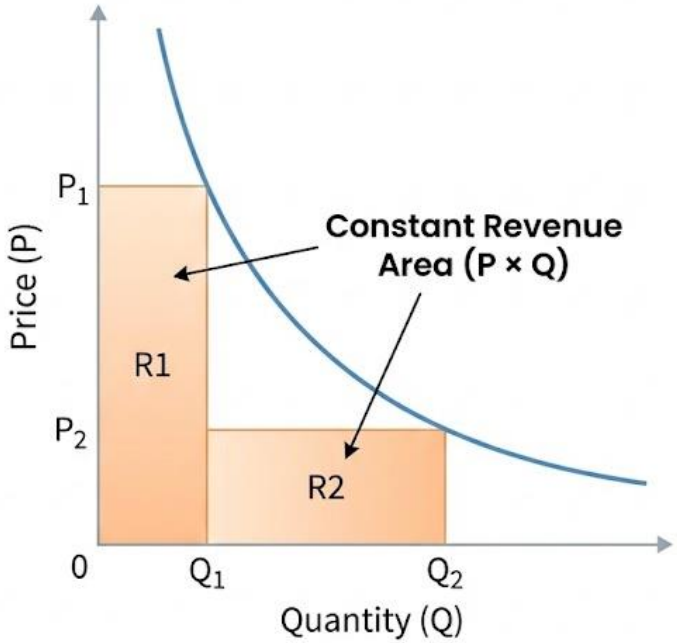
- In the tutorial, we analyze the Kaggle Avocado Prices dataset.
- **Data:** Weekly sales volume and average price.
- **Analysis:** We compare revenue during high-price weeks vs. low-price weeks.
- **Goal:** To determine if Avocado demand is Elastic or Inelastic.

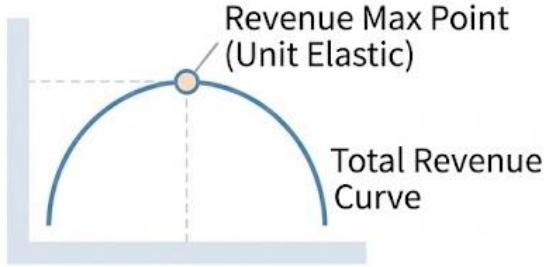


Unit Elasticity

- **Unit Elastic (|*PED*| = 1):**
  - % Change in Price = % Change in Quantity.
  - Revenue remains constant regardless of price.
- This is the theoretical maximum of the Revenue curve.





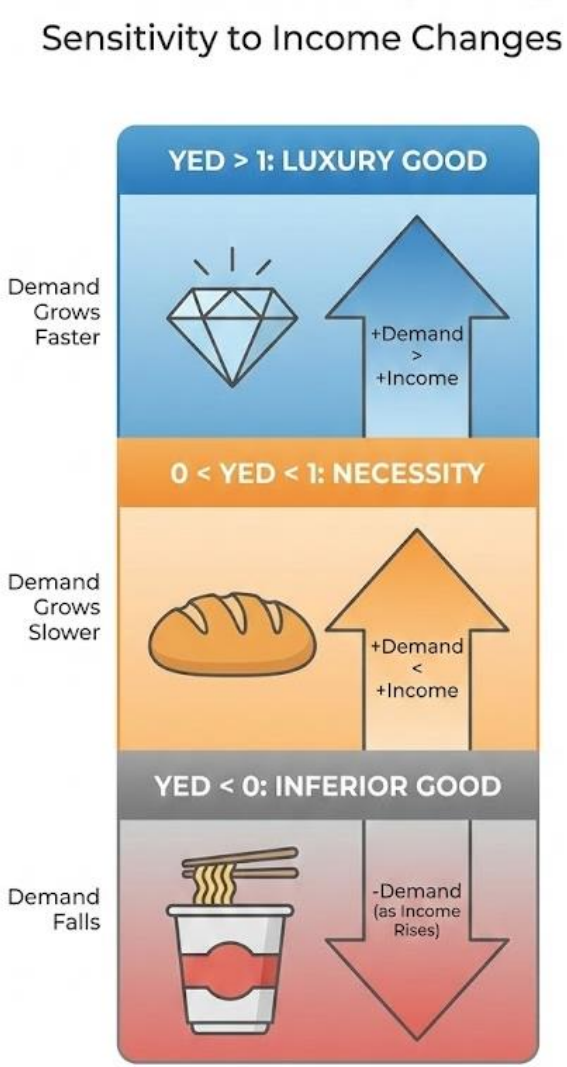


Revenue remains constant regardless of price changes.

https://economics.stackexchange.com/questions/36769/why-is-the-graph-of-unitary-elastic-demand-a-hyperbola

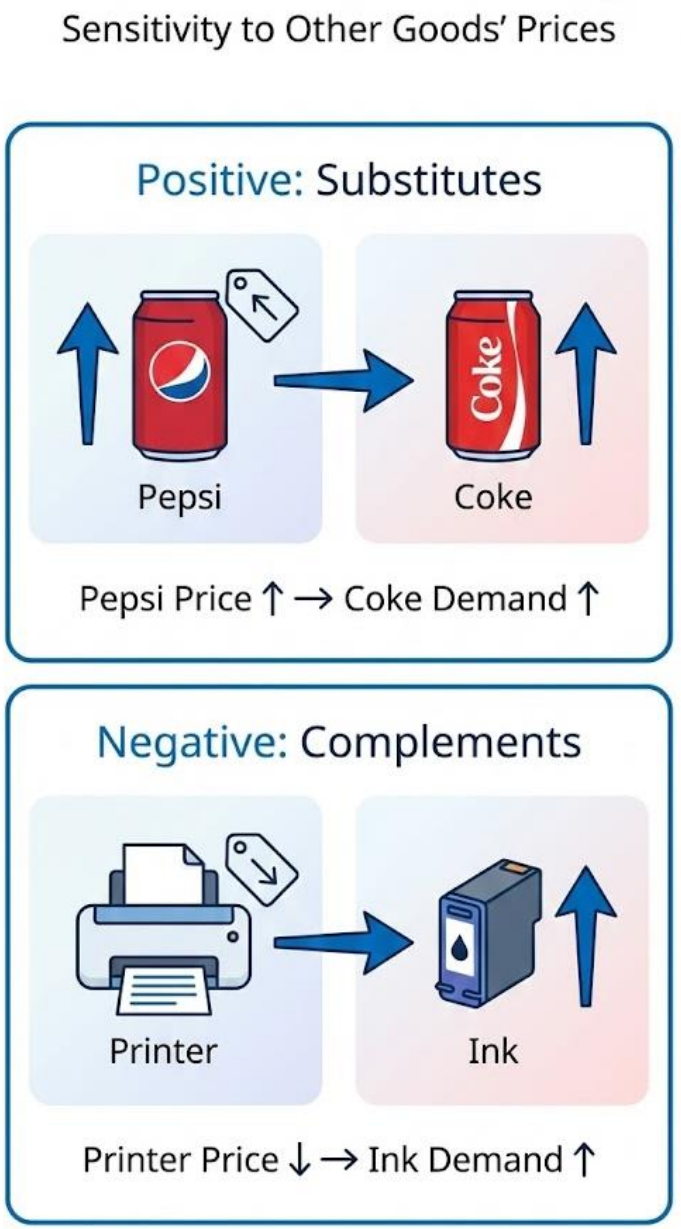
Income Elasticity of Demand

- Not just Price! We also measure sensitivity to **Income**.
- **Income Elasticity (YED):**
  - Positive = Normal Good.
  - Negative = Inferior Good.
  - > 1 = Luxury Good (Demand grows faster than income).



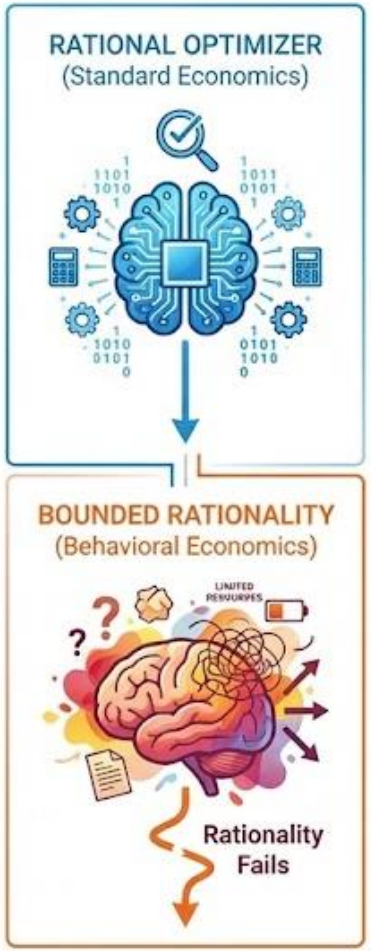
Cross-Price Elasticity

- Sensitivity to **Other Goods' Prices**.
- **Positive:** Substitutes (Pepsi price ↑ → Coke Demand ↑).
- **Negative:** Complements (Printer price ↓ → Ink Demand ↑).



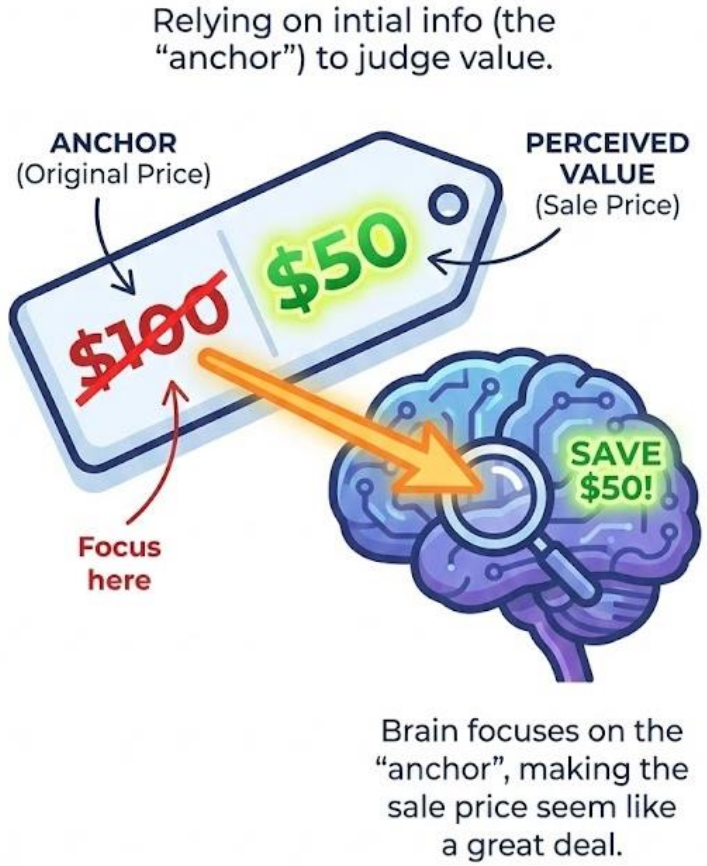
Behavioral Economics: When Rationality Fails

- Standard economics assumes we are "Rational Optimizers."
- **Behavioral Economics** challenges this.
- Humans have **Bounded Rationality**: We have limited cognitive power and information.



Heuristics and Biases

- We use shortcuts (**Heuristics**) to make decisions.
- **Anchoring:** We rely too heavily on the first piece of info (e.g., "Original Price $100" makes $50 look cheap).
- **Loss Aversion:** The pain of losing $10 > The joy of gaining $10.



The Decoy Effect (Marketing)

- **Scenario:**
  - Small Popcorn: $3
  - Large Popcorn: $7
- **Add Decoy:**
  - Medium Popcorn: $6.50
- **Result:** Everyone buys the Large because it looks like a "great deal" compared to the Medium.







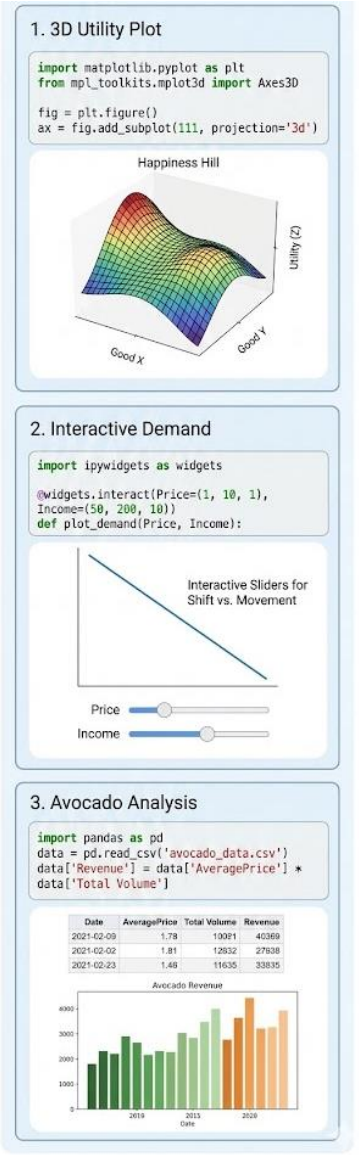
Part 3 Summary

- Elasticity = Sensitivity.
- Revenue Strategy depends on Elasticity (Raise price for Inelastic, Lower for Elastic).
- Humans are not perfectly rational; we are influenced by biases.
- **Next:** Let's apply this in Python.



Tutorial Overview

- **Goal:** Build a "Consumer Analytics" Dashboard.
- **Tools:** Python, Matplotlib, Pandas.
- **Key Modules:**
  - **3D Utility Plot:** Visualizing the happiness hill.
  - **Interactive Demand:** Sliders for Shift vs. Movement.
  - **Avocado Analysis:** Calculating revenue from real data.



Visualizing Utility in Python

- We will use mpl_toolkits.mplot3d.
- Function: plot_utility_3d(alpha, beta)
- You will adjust α and β to see how preferences change the shape of the hill.

Python Code Snippet

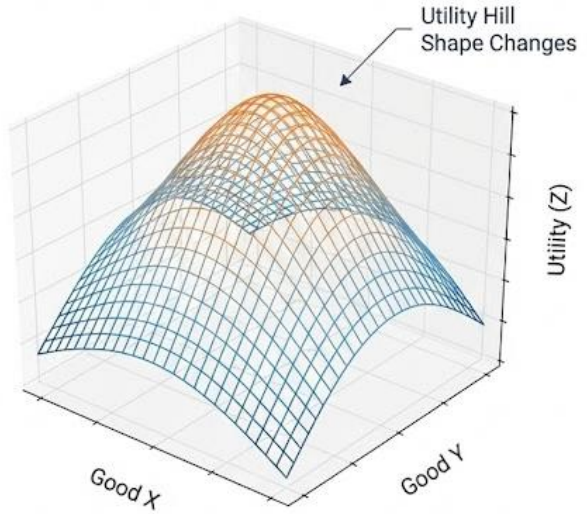
```
import mpl_toolkits.mplot3d

ax.plot_surface(X, Y, U, alpha=α, beta=β)

# Adjust α and β to see preference changes
```

Preference Parameters

Resulting 3D Wireframe Plot



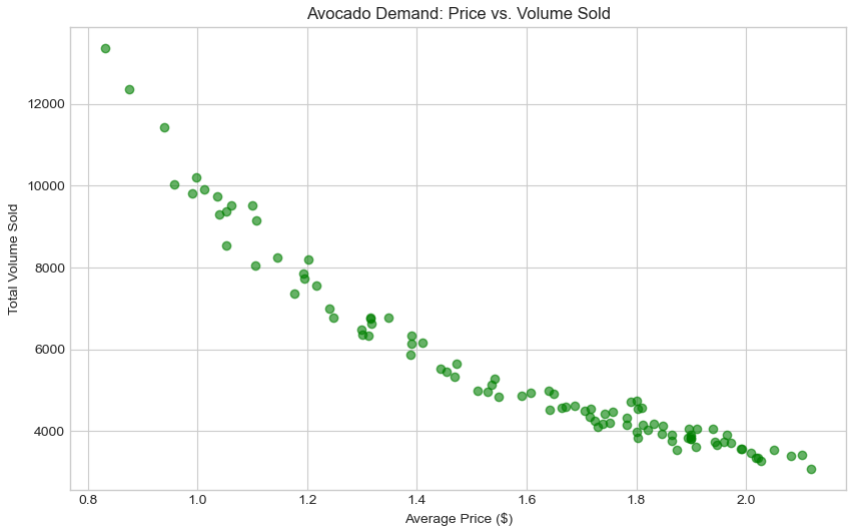
Interactive Demand Widgets

- We will use ipywidgets.
- You will control:
  - price (Red dot moves).
  - income (Blue line shifts).
  - substitute_price (Blue line shifts).
- **Task:** Observe the difference between changing Price vs. Income.



Homework Task 1: The Revenue Test

- **Context:** Synthetic Avocado Data (Generated in Notebook).
- **Task:**
  - Calculate Average Revenue when Price is High (>$Median).
  - Calculate Average Revenue when Price is Low (<$Median).
- **Question:** Based on which revenue is higher, is the demand Elastic or Inelastic?
- **Hint:** Remember the Price/Revenue relationship rules.



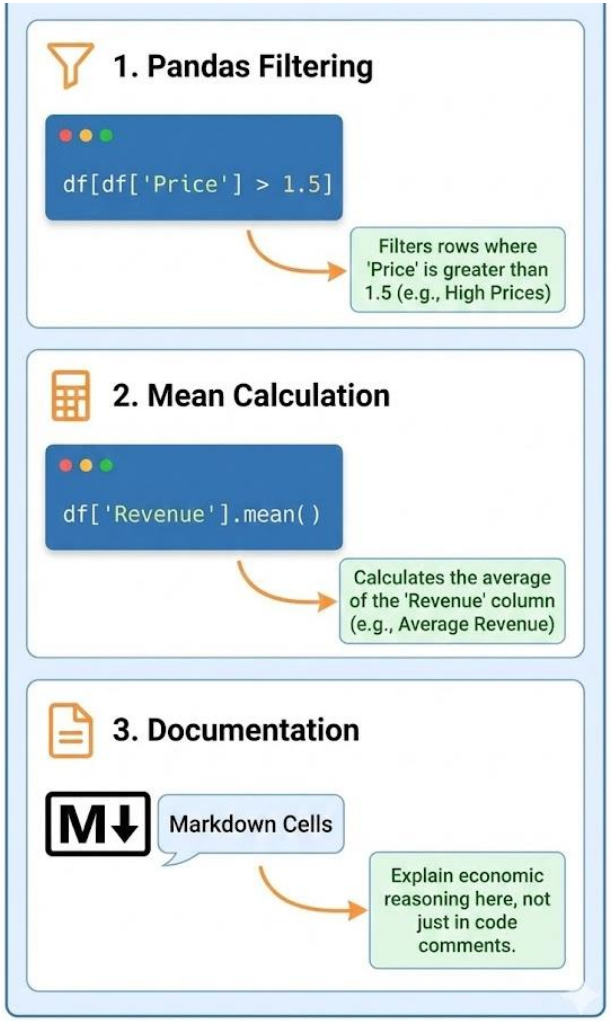
Homework Task 2: Daily Situation Analysis

- **Task:** Pick 3 goods you bought this week.
- **Classify:** Are they Elastic or Inelastic for YOU?
- **Mental Test:** "If the price doubled, would I stop buying it?"
- **Requirement:** Explain *why* (e.g., "Coffee is inelastic for me because I cannot function without it, and I dislike tea").



Python Tips for the Assignment

- **Pandas Filtering:**
  - `df[df['Price'] > 1.5]` to filter high prices.
- **Mean Calculation:**
  - `df['Revenue'].mean()` to get the average.
- **Documentation:** Use Markdown cells to explain your economic reasoning, not just code comments.



Conclusion

- Today we modeled the Consumer.
- We learned that **Value is Subjective** (Utility).
- We learned that **Responsiveness Varies** (Elasticity).
- **Next Week:** We move to the Supply Side.
  - **Session 3:** The Production Engine – Costs, Supply, and Profit Maximization.



Let’s check out our Notebook

